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Rafter Square Comprising a Heel Having an Integrated Tool Aperture

Abstract

A rafter square having an integrated tool aperture is disclosed. A tool aperture is formed in the heel of a rafter square which may be used to loosen or tight bolts in a tool such as on a circular saw for example. The heel may have a single, hexagonally shaped aperture to accommodate a hex bolt commonly found on circular saws in an embodiment. The tool aperture may be shaped in an elongated, stepped shape to accommodate multiple sizes of bolts.

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Background/Summary

BACKGROUND OF THE INVENTION
1. Field of the Invention
[0001] The present invention relates in general to rafter squares used in construction of homes and other structures to determine and mark cut angles for rafters and the like. More particularly, the

present invention is directed to rafter squares having a heel with an integrated tool aperture adapted for loosening and tightening mechanical fasteners.

2. Description of the Related Art

[0002] A rafter square is a multi-functional tool used for marking out cut lines on dimensional lumber, and can be used for framing a house, marking out and cutting rafters, and marking out and making cuts for decks and fences. As a result of their versatility, many carpenters and construction workers keep rafter squares on hand as part of their tools they take to the jobsite. In fact, many carpenters keep their rafter squares with them in their tool belts.

[0003] Carpenters who mark out and cut lines on dimensional lumber at a jobsite typically use a circular saw as well. Often, carpenters may have to replace a circular saw blade at the jobsite. Circular saw blades are replaced by using a wrench to engage with and loosen the hex bolt that secures the circular saw blade to the circular saw. However, this process may be time consuming for carpenters as they may have to leave the area of the jobsite in which they were working and walk to a remote area to find a proper wrench.

[0004] Accordingly, a need exists to provide rafter squares having multiple functions to allow carpenters to loosen and tighten bolts for circular or table saws at a jobsite.

SUMMARY OF THE INVENTION

[0005] In the first aspect, a rafter square is disclosed. The rafter square comprises an isosceles right triangular-shaped blade having a base side, a perpendicular side perpendicular to the base side, and a hypotenuse side that defines the hypotenuse between the base side and the perpendicular side. The blade has a plurality of threaded holes formed internally in the blade, each threaded hole having an opening at the base side of the blade.

[0006] The rafter square further comprises a heel of generally parallelepiped shape with an elongated body having a top face, a bottom face, a front edge, a back edge, and two side edges. The heel is coupled to the blade forming a right angle between the blade and the heel, the heel having one or more through-hole aperture for engaging with a fastening mechanism. The heel has a plurality of fastener through-holes positioned to align with the plurality of threaded holes of the blade when blade is attached to the heel.

[0007] The rafter square further comprises a plurality of fasteners for releasably attaching the heel to the blade where the top face is positioned immediately adjacent to base side of the blade, each fastener placed through each of the fastener through-holes of the heel and the corresponding threaded hole formed in the blade.

[0008] In a first preferred embodiment, the one or more through-hole apertures comprises a hexagonal-shaped aperture sized to releasable receive and engage with a hexagonal bolt. The aperture is preferably located on the heel proximal to the back edge of the heel. The one or more through-hole apertures preferably comprises an elongated and stepped aperture that provides multiple sized wrench apertures for use with different sized bolts. The one or more through-hole apertures preferably comprises a plurality of discrete apertures spaced apart on the heel. The blade preferably further comprises a blade notch positioned immediately above the aperture in the heel for receiving the head of the fastening mechanism. The rafter square is preferably sized to enable the engagement of a bolt securing a circular saw blade to a circular saw. The heel is preferably coupled to the blade via a plurality of fastening screws.

[0009] In a second aspect, a rafter square is disclosed. The rafter square comprises a triangular-shaped blade having a base side, a perpendicular side perpendicular to the base side, and a hypotenuse side that define the hypotenuse between the base side and the perpendicular side, and a heel of generally parallelepiped shape with an elongated body having a top face, a bottom face, a front edge, a back edge, and two side edges, the heel coupled to the blade forming a right angle between the blade and the heel, the heel having one or more through-hole apertures for engaging with a fastening mechanism.

[0010] In a second preferred embodiment, the one or more through-hole apertures comprises a hexagonal-shaped aperture sized to releasable receive and engage with a hexagonal bolt. The aperture is preferably located on the heel proximal to the vertex of the blade opposite the hypotenuse side. The one or more through-hole apertures preferably comprises an elongated and stepped aperture that provides multiple sized wrench apertures for use with different sized bolts. The one or more through-hole apertures preferably comprises a plurality of apertures spaced apart on the heel. The blade further preferably comprises a blade notch positioned immediately above the aperture in the heel for receiving the head of the fastening mechanism. The heel is preferably coupled to the blade via a plurality of fastening screws. The blade is preferably formed as a single piece. The rafter square preferably further comprises a level vial aligned parallel with the length of the heel providing visual indication of whether the rafter square is level.

[0011] In a third aspect, a rafter square is disclosed. The rafter square comprises a blade, and a heel of generally parallelepiped shape with an elongated body having a top face, a bottom face, a front edge, a back edge, and two side edges, the heel coupled to the blade forming a right angle between the blade and the heel, the heel having one or more through-hole apertures for engaging with a fastening mechanism.

[0012] In a third preferred embodiment, the one or more through-hole apertures comprises a hexagonal-shaped aperture sized to releasable receive and engage with a hexagonal bolt. The one or more through-hole apertures preferably comprises an elongated and stepped aperture that provides multiple sized wrench apertures for use with different sized bolts.

[0013] These and other features and advantages of the invention will become more apparent with a description of preferred embodiments in reference to the associated drawings.

Description

DESCRIPTION OF THE DRAWINGS

[0014] FIG. **1** is a perspective, exploded view showing a two-piece, rafter square having an integrated tool aperture in one or more embodiments.

[0015] FIG. **2** is a perspective view of the two-piece rafter square.

[0016] FIG. **3** is a perspective view of a rafter square having a level vial.

[0017] FIG. **4** is a perspective view of a one-piece rafter square in one or more embodiments.

[0018] FIG. **5** is a perspective view of the bottom of a one-piece rafter square having an aperture for accommodating multiple hexagonally shaped bolts.

[0019] FIG. **6** is a bottom view of a heel showing a single hexagonal aperture.

[0020] FIG. **7** is a bottom view of a heel showing an elongated, stepped, hexagonal aperture for accommodating multiple bolt sizes.

[0021] FIG. **8** is a bottom view of a heel showing multiple, discrete, hexagonal apertures spaced along the heel.

[0022] FIG. **9** is a perspective view of a rafter square having an aperture engaging with a bolt on a circular saw in one or more embodiments.

[0023] FIG. **10** is a perspective view of a rafter square having an aperture engaging with a bolt on for a generic tool or structure in one or more embodiments.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0024] Carpenters commonly will use a rafter square hand-in-hand with a circular saw, and often alternate between the use of the rafter square and the circular saw many times over the course of framing a house or laying out and cutting rafters. Carpenters frequently replace circular saw blades at the jobsite, such as to replace worn or damaged saw blades, or to replace saw blades with another type of saw blades having a different number of cutting teeth for finer or more aggressive cuts.

[0025] In order to replace a circular saw blade, carpenters conventionally will need to find the wrench for the size and shape of the bolt on the circular saw. In practice, this may require carpenters to leave the area at the jobsite in which they were working to find the correct wrench. In other cases, carpenters may have simply forgotten to bring their wrenches or encounter an unexpected bolt on a structure other than a circular saw that needs to be tightened. Alternatively, carpenters may carry the correct wrench in their toolbelts, but this practice can clutter a tool belt and make the tool belt more cumbersome.

[0026] In an embodiment, a rafter square having an integrated tool aperture (i.e., one or more through-hole apertures) is contemplated. The tool aperture is sized to receive and engage with a fastening mechanism such as bolts used in circular saws. A carpenter no longer needs to search for correct wrenches to remove or tighten circular saw bolts. Instead, carpenters simply reach for their rafter squares, a tool that they already carry, and use the tool aperture feature of the rafter square to remove or tighten the circular saw bolt. Hence having a tool aperture integrated into a rafter square saves time over the course of the construction project.

[0027] In addition, a rafter square with an integrated tool aperture may be useful when carpenters unexpectedly encounter bolts in structures or tools other than circular saws at jobsites which need to be tightened or loosened, and the carpenters do not have the proper wrenches in their possession. As most carpenters always carry rafter squares, they may simply use the rafter square to tighten or loosen bolts.

[0028] In an embodiment, the rafter square comprises a triangular-shaped blade, usually having cutout regions and markings indicating lengths and angles, and a heel that is attached to the bottom and extends beyond blade to form a “lip” or “fence” which can be placed against dimensional lumber to align the rafter square.

[0029] In an embodiment, a tool aperture is formed in the heel which a carpenter may use to loosen or tight bolts in a tool such as on a circular saw for example. In one or more embodiments, the heel may have a single, hexagonally shaped aperture to accommodate a hex bolt commonly found on circular saws in an embodiment. In an embodiment, the tool aperture may be shaped to engage with multiple sized bolts, such as in an elongated, stepped shape.

[0030] While embodiments described herein refer to specific examples such as circular saws or hexagonally shaped bolts (i.e., fastening mechanism), it shall be understood that the examples discussed and illustrated herein are for exemplary purposes only, and that other type of fastening mechanisms and other tools or structures which may benefit from embodiments described herein are contemplated in one or more embodiments.

[0031] FIG. **1** is a perspective, exploded view showing a two-piece, rafter square **101** in one or more embodiments. A right-handed coordinate system having mutually orthogonal x axis **10**, a y axis **12**, and a z axis **14** is also shown for reference.

[0032] A rafter square **101** comprises an isosceles right triangle shaped blade **140** and a heel **110**. The isosceles right triangle blade **140** has three vertices. The first vertex is labeled as vertex **160** (also labeled as “A”), the second vertex is labeled as vertex **162** (also labeled as “B”), and the third vertex is labeled as vertex **164** (also labeled as “C”). The vertex **160** may be referred to as the “pivot point” where the user can mark angles other than 90 degrees by pivoting the rafter square **101** relative to dimensional lumber. The angle formed by ∠CAB is 90 degrees forming a right angle, the angle formed by ∠ABC is 45 degrees, and the angle formed by ∠BCA is 45 degrees such that triangle ABC is an isosceles right triangle.

[0033] The isosceles right triangular shaped blade **140** has a base side **142** (formed by line segment AB parallel with the y axis **12**), a perpendicular side **144** (formed by line segment AC parallel with the z direction **14**) perpendicular to the base side **142**, and a hypotenuse side **146** (formed by line segment CB) that defines the hypotenuse **147** between the base side **142** and the perpendicular side **144**. The base side **142** and the perpendicular side **144** have a length “L.” The blade **140** has a top surface **170** and a bottom surface **172**. The blade **140** may have a plurality of cutout sections and length or angle indicia **182** such as a cutout **155***a*, a diagonal slot **155***b*, and a vertical slot **156** for example.

[0034] In an embodiment, a heel **110** has a generally parallelepiped, rectangular shape with an elongated body also having a length "L" parallel with the y axis **12**. The heel **110** has a top face **125**, a bottom face **126**, a front edge **127**, a back edge **128**, and two side edges **129***a* and **129***b*.
[0035] In an embodiment, the heel **110** is configured to be releasable attached to the blade **140**, so that the heel **110** may be interchangeable with other types of heels **110**. As shown in FIG. **1**, the blade **140** has a plurality of threaded holes **134***a*, **134***b*, **134***c*, **135***d*, and **135***e* formed internally in the blade **140** where each threaded hole **134***a*-**134***e* has a corresponding opening **135***a*, **135***b*, **135***c*, **135***d*, and **135***e* at the base side **142** of the blade **140**. The heel **110** has a plurality of fastener through-holes **130***a*, **130***b*, **130***c*, **130***d*, and **130***e* positioned to align with the plurality of threaded holes **134***a*-**134***e* of the blade **140** when blade **140** is attached to the heel **110**.
[0036] A plurality of fasteners **190** are employed to releasably attach the heel **110** to the blade **140**. When the top face **125** of the heel **110** is positioned immediately adjacent to base side **142** of the blade **140**, each fastener **190** is placed through each of the fastener through-holes **130***a*-**130***e* of the heel **110** and are threaded into the corresponding threaded holes **134***a*-**134***e* formed in the blade **140**. In other words, a first fastener **190** is placed through the fastener through hole **130***a* and into threaded hole **134***a*, a second fastener **190** is placed through hole **130***b* and into threaded hole **134***b*, a third fastener **190** is placed through hole **130***c* and into threaded hole **134***c*, a fourth fastener **190** is placed through hole **130***d* and into threaded hole **134***d*, and a fifth fastener **190** is placed through hole **130***e* and into threaded hole **134***e*.
[0037] When the heel **110** is attached to the blade **140**, a right angle is formed between the blade **140** and the heel **110**. As shown, the normal **150** for the blade is parallel with the x axis **10**, and the normal **120** of the heel **110** and the center line **119** of the normal of the aperture **112** is parallel with the z axis.
[0038] The heel **110** has a through-hole aperture **112** for engaging with a fastening mechanism **20**, which refers to a hexagonal bolt **20** for example. In an embodiment, the through-hole aperture **112** may be sized and shaped to receive and engage with a bolt **20** having a hexagonal head **22** that may be employed in a circular saw **30** for example (See FIGS. **6** and **9**). The aperture **112** Is a closed-end or box-end where the aperture **112** is placed around the bolt head **22** to completely encompass and surround the bolt **20**.
[0039] In an embodiment, the blade **140** further comprises a blade notch **180** positioned immediately above the aperture **112** in the heel **110** for receiving the bolt head **22** of the bolt **20** when the heel **110** is attached to the blade **140**. A blade notch **180** may provide additional clearance for the bolt head **22** of the bolt **20**, particularly when the thickness of the heel **110** "T" is less than the height of the bolt head **22**.
[0040] In an embodiment, the aperture **112** is located on the heel **110** proximal **122** to the back edge **128** of the heel **110**. When the rafter square **101** is assembled, the aperture **112** is positioned near the vertex **160** of the blade **240** opposite the hypotenuse side **146**, which may provide greater structural support to the blade **140** when a user is attempting to tighten or loosen a bolt **20**.
[0041] In other words, in an embodiment, a rafter square **101** comprises a triangular-shaped blade **140** and a heel **110**. The triangular-shaped blade **140** has a base side **142**, a perpendicular side **144** perpendicular to the base side **142**, and a hypotenuse side **146** that define the hypotenuse **147** between the base side **142** and the perpendicular side **144**. The heel **110** is of generally parallelepiped shape with an elongated body having a top face **125**, a bottom face, a front edge **126**, a back edge **127**, and two side edges **129***a* and **129***b*, the heel coupled to the blade forming a right angle between the blade and the heel, the heel **110** having one or more through-hole apertures **112** for engaging with a fastening mechanism **20** (i.e., hexagonal bolt **20** for example). In an embodiment, the heel **210** and the blade **240** are formed as a single piece **201** (See FIGS. **4** and **5**).
[0042] Stated in yet other words, in an embodiment, a rafter square **101** comprises a blade **140**, and a heel **110** of generally parallelepiped shape with an elongated body having a top face **125**, a bottom face, a front edge **126**, a back edge **127**, and two side edges **129***a* and **129***b*, the heel

coupled to the blade forming a right angle between the blade and the heel, the heel **110** having one or more through-hole apertures **112** for engaging with a fastening mechanism **20** (i.e., hexagonal bolt **20** for example).

[0043] FIG. **2** is a bottom, perspective view of the two-piece rafter square **101** having a single tool aperture **112**, where the heel **110** is attached to the blade **140** using the mechanical fasteners **190** such as screws placed through hole **130***a*. FIG. **2** also illustrates the region proximal **184** to the vertex **160** and the region distal **186** from the vertex **160**.

[0044] FIG. **3** is a bottom, perspective view of a rafter square **101***a* having a level vial **154**. A level vial window **152** is formed in the heel **110** to facilitate visibility of the level vial **154** during use.

[0045] FIG. **4** is a side, perspective view of a one-piece rafter square **201** in one or more embodiments. The rafter square **201** comprises a heel **210** and a blade **240** which are formed in one piece. Aperture **114** is formed onto heel **210**. FIG. **5** is a bottom, perspective view of a one-piece rafter square **201** having an aperture **114** accommodating multiple hexagonally shaped bolts.

[0046] FIGS. **6-8** illustrate the various shapes of through-hole apertures **112**, **112***a* and **114**. FIG. **6** is a bottom view of a heel **110**/**210** showing a single hexagonal through-hole aperture **112**. The through-hole aperture **112** may be sized and shaped to receive and engage with a bolt **20** having a hexagonal bolt head **22** that may be employed in a circular saw **30** for example. The aperture **112** Is a closed-end or box-end where the aperture **112** is placed around the bolt head **22** to completely encompass and surround the bolt head **22**. In an embodiment, the aperture **112** is positioned proximal to the pivot point **160**.

[0047] FIG. **7** is a bottom view of a heel showing an elongated, stepped, hexagonal aperture **114** for accommodating multiple bolt sizes, such as ⅝-inch, 9/16-inch, or ½-inch bolts for example. In an exemplary embodiment shown in FIG. **7**, the heel **110**/**210** has a through-hole aperture **114** for engaging with a plurality of bolts of differing sizes of bolt heads such as bolt head **22**, bolt head **24**, and bolt head **26** for example. The aperture **114** is similar to a closed-end or box-end wrench, where the aperture **114** is placed around the bolt head **22**, **24**, or **26** to receive the bolt head **22**, **24**, **26** and to completely encompass and surround the all of the bolt heads **22**, **24**, or **26**. Aperture **114** is a continuous aperture that surrounds all bolt sizes **22**/**24**/**26** without having partitions separating the sections for engaging with the multiple bolts.

[0048] In the exemplary embodiment depicted in FIG. **7**, the aperture **114** is contoured to engage with the largest bolt head **22** at the top of the aperture **114**, a smaller sized bolt head **24** near the middle of the aperture **114**, and the smallest bolt head **26** at the bottom of aperture **114**. To engage with the bolt head **22**/**24**/**26**, the user may laterally shift the position of the aperture **114** relative to the bolt head **22**/**24**/**26** until the bolt head **22**/**24**/**26** is engaged.

[0049] FIG. **8** is a bottom view of a heel **110**/**210** showing a multiple, discrete, hexagonal apertures **112** and **112***a* spaced along the heel. In an embodiment, aperture **112** is positioned proximal to the pivot point **184**, and aperture **112***a* is positioned distal from the pivot point **186**.

[0050] FIGS. **9** and **10** illustrate the use of a rafter square **101**/**201** in one or more embodiments. FIG. **9** is a perspective view of a rafter square **101**/**201** having an aperture **112** engaging with a bolt **20** on a circular saw **30** in one or more embodiments. FIG. **10** is a side view of a rafter square **101**/**201** having an aperture **112** engaging with a bolt **20** on a tool, piece of equipment. or structure.

[0051] Although the invention has been discussed with reference to specific embodiments, it is apparent and should be understood that the concept can be otherwise embodied to achieve the advantages discussed. The preferred embodiments above have been described primarily as a rafter square having an integrated tool aperture for tightening and loosening bolts. In this regard, the foregoing description of the rafter square having an integrated tool aperture is presented for purposes of illustration and description. It shall be apparent that other types of equipment would benefit from the aspects of the rafter square with an integrated tool aperture.

[0052] Furthermore, the description is not intended to limit the invention to the form disclosed herein. Accordingly, variants and modifications consistent with the following teachings, skill, and knowledge of the relevant art, are within the scope of the present invention. The embodiments described herein are further intended to explain modes known for practicing the invention disclosed herewith and to enable others skilled in the art to utilize the invention in equivalent, or alternative embodiments and with various modifications considered necessary by the particular application(s) or use(s) of the present invention.

Claims

1. A rafter square comprising: an isosceles right triangular-shaped blade having a base side, a perpendicular side perpendicular to the base side; and a hypotenuse side that defines the hypotenuse between the base side and the perpendicular side, the blade having a plurality of threaded holes formed internally in the blade, each threaded hole having an opening at the base side of the blade; a heel of generally parallelepiped shape with an elongated body having a top face, a bottom face, a front edge, a back edge, and two side edges, the heel coupled to the blade forming a right angle between the blade and the heel, the heel having one or more through-hole apertures for engaging with a fastening mechanism, the heel having a plurality of fastener through-holes positioned to align with the plurality of threaded holes of the blade when blade is attached to the heel; and, a plurality of fasteners for releasably attaching the heel to the blade where the top face is positioned immediately adjacent to base side of the blade, each fastener placed through each of the fastener through-holes of the heel and the corresponding threaded hole formed in the blade.
2. The rafter square of claim 1, wherein: the fastening mechanism comprises a hexagonal bolt; and the one or more through-hole apertures comprises a hexagonal-shaped aperture sized to releasable receive and engage with the hexagonal bolt.
3. The rafter square of claim 2, wherein the aperture is located on the heel proximal to the back edge of the heel.
4. The rafter square of claim 1, wherein: the fastening mechanism comprises a hexagonal bolt; and the one or more through-hole apertures comprises an elongated and stepped continuous aperture to receive and engage with different sized hexagonal bolts.
5. The rafter square of claim 1, wherein the one or more through-hole apertures comprises a plurality of discrete apertures spaced apart on the heel.
6. The rafter square of claim 1, wherein the blade further comprises a blade notch positioned immediately above the aperture in the heel for receiving a head of the fastening mechanism.
7. The rafter square of claim 1, wherein the rafter square is sized to enable the engagement of a bolt securing a circular saw blade to a circular saw.
8. The rafter square of claim 1, where the heel is coupled to the blade via a plurality of fastening screws.
9. A rafter square comprising: a triangular-shaped blade having a base side, a perpendicular side perpendicular to the base side, and a hypotenuse side that define the hypotenuse between the base side and the perpendicular side; and, a heel of generally parallelepiped shape with an elongated body having a top face, a bottom face, a front edge, a back edge, and two side edges, the heel coupled to the blade forming a right angle between the blade and the heel, the heel having one or more through-hole apertures for engaging with a fastening mechanism.
10. The rafter square of claim 9, wherein: the fastening mechanism comprises a hexagonal bolt; and the one or more through-hole apertures comprises a hexagonal-shaped aperture sized to releasable receive and engage with the hexagonal bolt.
11. The rafter square of claim 10, wherein the aperture is located on the heel proximal to the vertex of the blade opposite the hypotenuse side.
12. The rafter square of claim 9, wherein: the fastening mechanism comprises a hexagonal bolt; and the one or more through-hole apertures comprises an elongated and stepped continuous aperture to receive and engage with different sized hexagonal bolts.

13. The rafter square of claim 9, wherein the one or more through-hole apertures comprises a plurality of apertures spaced apart on the heel.
14. The rafter square of claim 10, wherein the blade further comprises a blade notch positioned immediately above the aperture in the heel for receiving the head of the fastening mechanism.
15. The rafter square **101** of claim 9, where the heel is coupled to the blade via a plurality of fastening screws.
16. The rafter square of claim 9, wherein the heel and the blade are formed as a single piece.
17. The rafter square of claim 9, further comprising a level vial aligned parallel with the length of the heel providing visual indication of whether the rafter square is level.
18. A rafter square comprising: a blade; and, a heel of generally parallelepiped shape with an elongated body having a top face, a bottom face, a front edge, a back edge, and two side edges, the heel coupled to the blade forming a right angle between the blade and the heel, the heel having one or more through-hole apertures for engaging with a fastening mechanism.
19. The rafter square of claim 18, wherein the fastening mechanism comprises a hexagonal bolt; and the one or more through-hole apertures comprises hexagonal-shaped aperture sized to releasable receive and engage with a hexagonal bolt.
20. The rafter square of claim 18, wherein the fastening mechanism comprises a hexagonal bolt; and the one or more through-hole apertures comprises an elongated and stepped continuous aperture to receive and engage with different sized hexagonal bolts.